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An improved method of extreme value analysis

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ABSTRACT

An improved method to estimate the probability of extreme events from independent observations is presented. The method, called VWLS, is based on minimizing the variance of order-ranked observations plotted according to their true probability and applying the least squares fitting. We show by Monte-Carlo simulations that this method provides estimates for the extremes that are considerably better than obtained by presently available EVA methods, particularly for small data sets. An additional benefit of VWLS is that its application requires no subjective methodological decisions by the user.
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1. Introduction

The problem of how to estimate the probability of very rare events by observed data is crucial in hydrology, climatology, structural engineering, reinsurance business and many other sectors concerned with hazards caused by extreme events. The field of extreme value analysis (EVA) has a more than a century-long history, and is quite advanced both in terms of its mathematical foundations and implementations to various applications, such as flood frequency analysis. For performing EVA, numerous different methods exist and are discussed in textbooks and review articles (e.g. Dixon, 1950; Gumbel, 1958; Pickands, 1975; Castillo, 1988; Smith, 1989; Hosking and Wallis, 1997; Coles, 2001; Beirlant et al., 2004; Castillo et al., 2005; Jordaan, 2005; McNeil et al., 2005; Reiss and Thomas, 2007). There exist several commercial and open software packages for the purposes of EVA (Stephenson and Gilleland, 2005; Gilleland et al., 2013).

The advantage of the extreme value theory is that, ideally, the original parent distribution and its cumulative distribution function need not to be known, because the distribution of the extremes asymptotically approaches a known distribution. Historically, (Fisher and Tippett, 1928; Gumbel, 1958) the extreme value theory has been utilized so that the extremes are assumed to be distributed according to one of the theoretical asymptotic extreme value distributions, Weibull, Fréchet or Gumbel, depending on an *a priori* assumption of the parent distribution. Nowadays, these distributions are usually combined into the Generalized Extreme Value distribution (GEV) with a probability density function (PDF) as

$$F(x) = \exp\left[-(1+\gamma(x-\mu)/\alpha)^{-1/\gamma}\right] \tag{1}$$

where $\alpha > 0$ and $1 + \gamma(x-\mu)/\alpha > 0$. GEV is a three parameter distribution, the limiting value of which at $\gamma = 0$ is the Gumbel distribution

$$F(x) = \exp(-\exp(-y)) \tag{2}$$

where y is the reduced variate

$$y = (x-\mu)/\alpha \tag{3}$$

Here, μ is the location parameter and α is the scale parameter. The GEV distribution has a finite upper tail for the shape parameter $\gamma < 0$ (Weibull), whereas for $\gamma > 0$ (Fréchet) and $\gamma = 0$ (Gumbel) there is no upper bound.

The fit to estimate the parameters μ and α is traditionally performed on a Gumbel probability graph, where the ordinate is the reduced variate from the estimated $F(x)$, plotted as $y = -ln(-ln(F(x)))$, and the abscissa is the variate value x. This plot transforms the Gumbel distribution model into a line with a slope of $1/\alpha$ and intercept μ/α. Once the fit is made and the parameters μ and α are determined, one can calculate x that corresponds to any probability $F(x) = P$ and the corresponding return period R. Often, one will need to estimate extreme values of X that are higher than those included in the data. Graphically this corresponds to extrapolation along the fit on the graph (Fig. 1). The modern method is to use the GEV distribution (Eq. (1)) and apply some fitting procedure.

However, there is no universally accepted method of EVA. In contrast, different authors and software provide different methods

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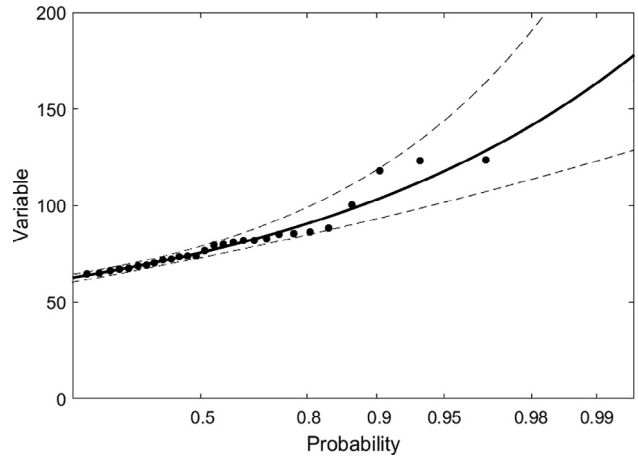

Fig. 1. Illustration of EVA. Extreme data are here plotted on Gumbel paper, where the double-logarithm of the non-exceedance probability is on the abscissa and the variable value on the ordinate. An estimate of the underlying cumulative distribution function and its confidence intervals are also shown.

and often propose several alternative means for the related procedures. Thus, in practice, performing EVA involves considerable subjectivity. EVA is typically performed either by using the Maximum Likelihood Method (MLE) or the Probability Weighted Moments method (PWM), or by associating each observation to its probability and making a fit. In the latter case, one needs to select

- The *a priori* functional form of the cumulative distribution function (CDF) of the extreme value data
- The plotting formula
- The method of weighing the data
- The fitting method

Three theoretically based extreme value distributions exist for the CDF, fifteen plotting formula are proposed in the literature and seven methods are commonly used for the weighing of data. The fitting can be done with respect to either probability or the variable itself on a probability plot. Bayesian methods are also in use. Consequently, there are around 1000 ways to perform EVA. This means that the results of EVA presented in the literature, and used for important decision making, are contaminated with significant methodological uncertainties. As demonstrated earlier (Makkonen, 2006; 2008a) this may result in serious underestimation of risks when the observed data sets are small.

Here, an improved and more objective method of EVA is presented. In this method, none of the selections, discussed above, need to be made by the user. Instead, a three parameter Generalized Extreme Value Distribution (GEV) is applied, the plotting is based on the true rank probabilities, and the weighing and fitting are linked to each other by solving their optimal selection iteratively. We then shown by Monte-Carlo simulations that this method outperforms the other widely used methods of EVA, including the MLE and PWM, particularly for small data sets. Finally, the benefits and weaknesses of this novel method are discussed.

2. The EVA method

2.1. The fitted distribution

In the method proposed here, GEV is used and all its three parameters are varied. It should be emphasized that the use of GEV here is not based merely on an assumption that the observations are extremes from an infinite set of data, i.e. on full convergence towards the asymptotic extreme value distribution. Rather, this selection is based on that, even in the case of incomplete convergence, i.e. the so-called penultimate distribution, extreme value data can be fitted by a three-parameter GEV very well (Furrer and Katz, 2008). Thus, in principle, the method is applicable to any type of iid block extreme data.

2.2. The plotting positions

The cumulative distribution function of N number of annual maxima is estimated from the order statistics so that each maximum is assigned a rank m in ascending value $1 \leq m \leq N$. For a conventional analysis of order ranked extremes, one needs to associate them to some probability, the so-called plotting position. This is done here by the formula

$$P_m = m/(N+1) \tag{4}$$

It has been shown (Makkonen, 2006; 2008a,b; Makkonen et al., 2013) that this, so called Weibull plotting position, provides the correct rank probability for any distribution of extremes.

2.3. Weighing and fitting

There is significant controversy related to the methods of fitting a distribution to extreme value data (Katz et al., 2002; Makkonen, 2008a). For example, the performance of the maximum likelihood method for the GEV distribution may be erratic for small samples (Martins and Stedinger, 2000) and the method of probability weighed moments is not optimal for the Gumbel-distributed samples (Rasmussen and Gautam, 2003). Our fitting method deviates from the conventional methods and is described below.

In the conventional fitting methods, the error, i.e. the difference between the data and the model output, is minimized with respect to the probability P. In contrast, in our method the differences are minimized with respect to the variate X. This is motivated by the fact that, in the case of order ranked data, P is not a variate (Makkonen and Pajari, 2014) and that Monte-Carlo simulations support this choice (Pajari et al., 2019). This choice of the axis is one of the key elements in the success of the method demonstrated in this paper.

There are number of ways to fit a distribution to the plotted data. In addition to the simple method of least squares, methods that weigh each extreme based to its statistical confidence have been proposed (Lieblein, 1974; Landwehr et al., 1979; Hosking et al., 1985; Wang, 1991; Castillo, 1988; Katz et al., 2002). It is noteworthy that all curve fitting techniques implicitly involve giving weights to the data points.

In EVA, the issue of the weights is particularly important and controversial. Giving larger weights to the most extreme observations may be supported by the possibility that the highest extremes do not belong to the same population with more moderate extremes. Thus, putting more weight on the points with a low probability would seem to be necessary for the phenomena whose formation mechanisms are poorly understood. However, this line of thinking contradicts the commonly used approach (Harris, 1996; Landwehr et al., 1979; Hosking et al., 1985) in which the points are weighted based on their confidence, thus effectively neglecting the data at the tail of the distribution.

In our method, the weighted least squares are used in making the fit to GEV. As mentioned above, in our fitting method, the errors are minimized with respect to the random variable X, and not P. For this purpose, ξ_m is solved from the GEV distribution (Eq. (1)) as

$$\xi_m = F^{-1}[m/(N+1)] = \mu + \frac{\alpha}{\gamma}\left[-1 + (-\ln F(\xi_m))^{-\gamma}\right] \tag{5}$$

Here, ξ_m is the value of X given by the estimate of the cumulative distribution function F at the probability corresponding to the observation x_m.

In the method, the weights w_m for the data points are given according to the inverse of the variance of ξ_m. The variance can be calculated when the cumulative distribution function at $F(\xi_m)$ is known. This is explained below.

Consider the stochastic variable X_m from the order ranked data.

$$x_1 \leqslant x_2 \leqslant \ldots \leqslant x_N \tag{6}$$

There are $m-1$ independent stochastic variables less than and $N-m$ larger than X_m. Hence, the probability density function $f_m(\xi_m)$ of X_m can be determined by (e.g. Castillo, 1988)

$$f_m(\xi_m) = \frac{N!}{(m-1)!(N-m)!} F(\xi_m)^{m-1}[1-F(\xi_m)]^{N-m} f(\xi_m) \tag{7}$$

where m is rank number, N is the total number of variables, $F(\xi_m)$ is the cumulative distribution function and $f(\xi_m)$ is the density function. For the GEV-distribution, $F(\xi_m)$ is given by Eq. (1) and

$$f(x) = \frac{1}{\alpha}\left[1+\gamma\left(\frac{x-\mu}{\alpha}\right)\right]^{(-1/\gamma)-1} \exp\left[-\left[1+\gamma\left(\frac{x-\mu}{\alpha}\right)\right]^{-1/\gamma}\right] \tag{8}$$

for $x > \mu - \alpha/\gamma$ when $\gamma > 0$ and for $x < \mu + \alpha/(-\gamma)$ when $\gamma < 0$. Outside this range the density is zero. When $\gamma = 0$, the density function is defined as

$$f(x) = \frac{1}{\alpha}\exp\left[-\left(\frac{x-\mu}{\alpha}\right)\right]\exp\left\{-\exp\left[-\left(\frac{x-\mu}{\alpha}\right)\right]\right\} \tag{9}$$

Hence, mean of the variable ξ_m are numerically calculated from the distribution function by

$$\bar{\xi}_m = \int_{-\infty}^{\infty} \xi_m f_m(\xi_m) d\xi_m \tag{10}$$

and the variance σ_m^2 by

$$\sigma_m^2 = \int_{-\infty}^{\infty} \left(\xi_m - \bar{\xi}_m\right)^2 f_m(\xi_m) d\xi_m \tag{11}$$

The weights w_m are then determined by equation

$$w_m = \frac{1}{\sigma_m^2} \tag{12}$$

and are normalized so that

$$\sum_{m=k}^{N} w_m = 1 \tag{13}$$

where k is the index of smallest value of x_k that is taken into account in the fitting, as explained below.

Using these weights, the sum

$$\sum_{m=k}^{N} w_m (x_m - \xi_m)^2 \tag{14}$$

is minimized.

Thus, in this method, an estimate of the cumulative distribution function $F(\xi)$ needs to be applied when determining the weights. On the other hand, the weights are required in determining an estimate of $F(\xi)$. This situation is solved by considering both the weights and the fitted function simultaneously in an iteration process, as explained in the following.

First, equal weights are given for all the extremes for obtaining the first estimate of the distribution function and the weights for the second run. At each iteration step, the weights are adjusted according to Eqs. (11)–(14) and the three parameters of GEV are found by minimizing the sum in Eq. (14). When the estimated cumulative distribution function $F(\xi)$ and the weights w_m do not change anymore in the iteration, the CDF $F(\xi)$ is accepted as the final result. Typically, 10–20 iteration steps are required. While testing the method excessively (for example by 44,800 natural samples in Fig. 4), no data sets were found for which the method would not converge.

In some cases, the variable X may have an absolute lower limit according to the fitted GEV. This can sometimes be seen also in real data, for example the amount of snow on the ground, for which the limiting value is zero. Since, in EVA, we are interested in the upper tail of the distribution, values at the lower limit are not useful. Therefore, a criterion on how close to the theoretical lower limit the smallest observed extremes are, is applied, and they are rejected from the analysis accordingly. This is explained in detail in the following.

The smallest extremes are excluded in our analysis, i.e. they are given zero weights, if they lay very close to the lower asymptotic limit given by the estimated $F(\xi)$. This is done based on the criterion which compares the difference between the smallest and the second smallest ξ_m value, ξ_k and ξ_{k+1} respectively, and the difference between the largest value ξ_N and the smallest value ξ_k. If

$$\frac{\xi_{k+1} - \xi_k}{\xi_N - \xi_k} < b \tag{15}$$

then the smallest observation is eliminated from the analysis and only observations from x_{k+1} to x_N are used in the new iteration process. The elimination process continues to the next smallest extreme until the criterion in Eq. (15) is not met. If there are no more than five observations left for the analysis, then the data are considered unreliable and rejected. Here, $b = 0.001$ is used in Eq. (15).

2.4. Confidence intervals

When using this method, for each rank, confidence intervals can be determined by calculating the quantile of interest from Eq. (7). These rank intervals can be transferred from the probability scale to the variable scale. As an example, Fig. 1 shows a GEV distribution that has been fitted to the rank confidence intervals, which have been interpolated between each rank and extrapolated. Here, the standard deviation of the m-th largest order statistic has been used as a confidence level of interest.

3. Performance of the VWLS method

3.1. Evaluation of a fitting method

We call our improved EVA method described above, VWLS, since it combines minimization of the variance in x (V), Weibull plotting positions (W) and least squares (LS). For demonstrating the power of VWLS, we investigate in the following how well different methods estimate the true CDF of extremes. To compare the methods, a procedure described in Makkonen et al. (2013) and Pajari et al. (2019) is used, instead of the conventional comparison of the statistics of the distribution parameters or the quantiles. The compelling reasons for this are outlined in the following.

The distribution parameters γ, μ and α are non-linearly related to $P = F(x)$. Therefore, there is no reason to believe that e.g. the bias of the estimated distribution parameters can be used as a measure of a good fit for the distribution itself. Another way to express this argument is to note that a non-linear transformation of an unbiased estimator is biased. Consequently, contrary to the commonly applied intuition, good estimates of the distribution parameters do not necessarily imply a good fit of the distribution (see Pajari et al., 2019).

We demonstrate the misleading nature of assessing the goodness of fit by the distribution parameters by a simple example. In Fig. 2 the tail of a Gumbel distribution with $\mu = 0$, $\alpha = 1$ is shown as the solid line, and two estimates for it, one with $\mu = 0.05$, $\alpha = 1.045$ (blue dashed line), and another with $\mu = 0.1$, $\alpha = 0.95$ (red dashed line) are also shown. Both parameters, μ and α, are closer to the correct values for the blue line. Nevertheless, the red line provides much better estimates for the probability of the extreme values.

When the distribution includes several parameters, three in the case of GEV, the use of the goodness of the parameters in evaluating the goodness of the fit is even more misleading. In such a case, the properties of a single parameter do not tell anything of the goodness of fit in terms of e.g. quantiles, and a situation may arise where errors in the parameters cancel out in some range of values, but not in others.

Another popular method has been to use the bias or RMS error of the quantiles as a measure of goodness of fit. However, for distributions, such as GEV, also the quantiles are non-linearly related to the probability. Thus, e.g. the bias of a quantile is an inappropriate measure in probabilistic sense (Pajari et al., 2019). Accordingly, the goodness of fit in the EVA methods need to be compared not by the statistics of the parameters γ, μ, α, nor the quantiles, but by the property for which the CDF estimates are *defined*, i.e. the probability. This can be assessed by considering the frequency of samples falling into equally spaced probability intervals (Makkonen et al., 2013, Pajari et al., 2019). This is explained in the following.

Assume a variate X and its cumulative distribution function F. Divide the probability axis in $p+1$ equal intervals by values $i/(p+1), i = 0, ..., p+1$, and calculate values $x_{(i)} = F^{-1}(i/(p+1))$. By definition, for an arbitrary observation x the probability of falling to bin $(x_{(i-1)}, x_{(i)}]$ equals $1/(p+1)$ for $i = 1, ..., p+1$ and for all positive integers p. Hence, with increasing number of observations, the relative frequency of observations x falling to bin $(x_{(i-1)}, x_{(i)}]$ approaches the probability $1/(p+1)$. This fundamental property can be used when comparing fitting methods.

The appropriate evaluation method can be outlined by the following steps.

1. Choose a CDF, $F(x)$.
2. Take a sample including N values of X, which are $x_1, ..., x_N$ in ascending order.
3. Using this sample and the different fitting methods, find the respective estimated CDF, $F_1(x), F_2(x), \ ...$
4. Take the next random value x_{N+1} from the population of X.
5. Record for each function $F_j(x)$ the bin $\left[F_j^{-1}((r-1)/(N+1)), F_j^{-1}(r/(N+1))\right]$ to which x_{N+1} belongs.
6. Repeat, n times, steps 2–5 until the relative frequency $n_{j,r}/n$ (=number of x_{n+1}-values falling in bin r divided by n) no more changes.
7. The goodness of the fitting procedure can now be evaluated using the root squared sum C as

$$C = \sqrt{\sum_{r=1}^{N+1} \left(\frac{n_{j,r}}{n} - \frac{1}{N+1} \right)^2} \tag{16}$$

3.2. Comparison of EVA methods

Monte Carlo simulations with 100,000 rounds to calculate the bin frequency and C were made by fitting the data using the Maximum Likelihood Method (Fisher, 1922; Hosking, 1985), the Probability Weighted Moments (Greenwood et al., 1979; Hosking et al., 1985), L-moments (Hosking, 1990; Hosking and Wallis, 1997) and LH2 moments (Wang, 1997), as well as the VWLS method presented here. The random data were produced from Gumbel, Fréchet and Weibull type distributions for demonstrating the method in case of extreme value distributions. The parameters of the distributions were selected so that they represent typical values in environmental data. An example of the distribution of samples into the bins in these simulations is shown in Fig. 3.

The results of C for Fréchet distribution with parameters $\gamma = 0.3$, $\alpha = 10$ and $\mu = 70$ are shown in Table 1. The columns of the table show C for simulations by different methods. The 'Ideal' column presents C with respect to the actual distribution in question. This value is calculated by replacing the fitted CDF $F_j(x)$ in step 5 by the actual chosen CDF, $F(x)$. Similar results for the reversed Weibull distribution with parameters $\gamma = -0.3$, $\alpha = 10$ and $\mu = 70$ are shown in Table 2, and for the Gumbel distribution with parameters $\alpha = 10$ and $\mu = 70$ in Table 3. However, it is noteworthy that, for a fixed shape parameter, a GEV distribution is

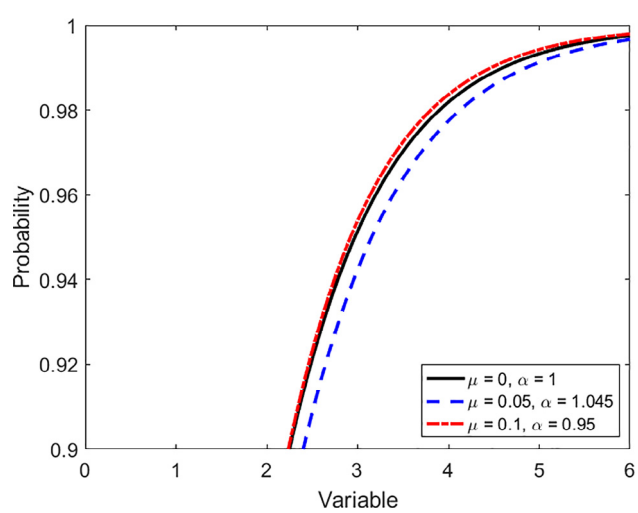

Fig. 2. Example showing how less biased distribution parameters may lead to a more biased estimate of the probability of extreme events. Gumbel distribution with $\mu = 0$, $\alpha = 1$ is shown as the solid black line, and two estimates for it, one with $\mu = 0.05$, $\alpha = 1.045$ (blue line), and another with $\mu = 0.1$, $\alpha = 0.95$ (red line) are also shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

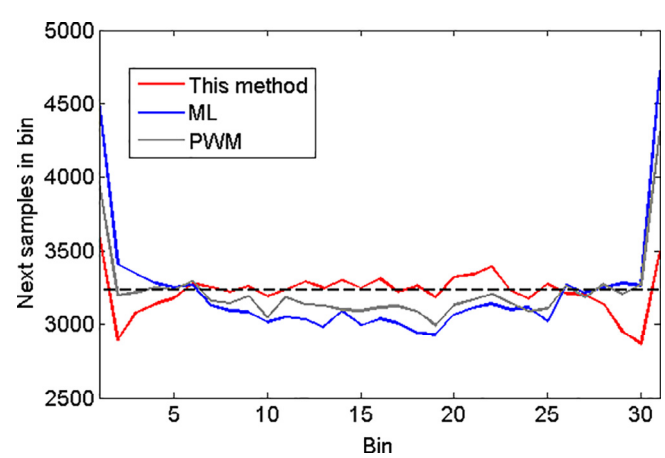

Fig. 3. Distribution of next mth samples into the bins that correspond to the mth order rank in a Monte-Carlo simulation of 100,000 data sets randomly generated from a Gumbel distribution with $\alpha = 10$ and $\mu = 70$.

Table 1
Comparisons for the Fréchet type distribution with parameters $\gamma = 0.3$, $\alpha = 10$ and $\mu = 70$. The columns show $C \cdot 10^3$ for VWLS, Maximum Likelihood, Probability Weighted Moments, L-moments and LH2 moments methods. The 'ideal' column presents $C \cdot 10^3$ when compared with the actual distribution.

N	Ideal	VWLS	ML	PWM	L	LH2
10	3	17	63	49	49	76
15	4	14	44	35	35	55
20	3	12	31	24	24	42
30	3	7	19	16	16	31
50	3	7	10	10	10	23
100	3	4	5	5	5	15

Table 2
Comparisons for the reversed Weibull type distribution with parameters $\gamma = -0.3$, $\alpha = 10$ and $\mu = 70$. The columns show $C \cdot 10^3$ for VWLS, Maximum Likelihood, Probability Weighted Moments, L-moments and LH2 moments methods. The 'ideal' column presents $C \cdot 10^3$ when compared with the actual distribution.

N	Ideal	VWLS	ML	PWM	L	LH2
10	3	9	70	39	39	56
15	3	11	46	23	23	36
20	3	10	34	18	19	29
30	3	8	23	13	13	21
50	3	5	13	8	8	12
100	3	4	6	5	5	6

Table 3
Comparisons for the Gumbel type distribution with parameters $\alpha = 10$ and $\mu = 70$. The columns show $C \cdot 10^3$ for VWLS, Maximum Likelihood, Probability Weighted Moments, L-moments and LH2 moments methods. The 'ideal' column presents $C \cdot 10^3$ when compared with the actual distribution.

N	Ideal	VWLS	ML	PWM	L	LH2
10	5	11	65	44	44	64
15	2	12	43	27	27	43
20	3	10	33	21	21	35
30	3	8	21	13	13	25
50	3	6	11	8	8	15
100	3	5	5	5	5	9

location-scale invariant, so that the results are the same as given in Tables 1–3 when, say, $\mu = 0$ and $\alpha = 1$.

The results in Tables 1–3 show that, for data simulated from the three types of extreme value distributions, the method; VWLS, presented here, outperforms the Maximum Likelihood and Probability Weighted moments clearly up to $N = 100$. It also outperforms the L-moments method, for which the results are equal to PWM, and the LH2 moments method. As shown in Fig. 3, for small data sets, VWLS is better particularly in predicting the probability of the largest and smallest value, which are of major interest in e.g. flood frequency analysis.

3.3. Estimation by extrapolation

The typical use of EVA in flood frequency analysis is to extrapolate to a probability lower than those covered by the observed data of extremes. For testing our new EVA method, it would be preferable to compare the methods in extrapolation against the selected CDFs directly. This cannot be done by the mean of the simulated values of X_m, where m refers to a fixed rank, because $\left(\bar{x}_m, P_m\right)$ does not fall on a non-linear CDF. Hence, theoretically, the best method would be to compare the methods similarly to what was done in Section 3.1, i.e. by fixing a value of x and investigating how well $P(x)$ will be predicted in the mean.

However, there are two problems with this procedure. First, extrapolation to a non-exceedance probability at a fixed x results, in a Monte Carlo simulation, to occasional fits with $P = 1$. This causes a methodological problem in comparing the mean of $P(x)$ with the $P = F(x)$ from the CDF. Second, there is a practical requirement that in most international regulating codes and standards EVA is applied in such way that a return value $x(P)$, that corresponds to a given probability P, is estimated, not *vice versa*. This value of x is the basic value in standards, such as the Eurocode and is, therefore, of outmost interest in e.g. engineering design.

For these reasons, we compare here the EVA methods in extrapolation by using the variable value $x_m = x(P_m)$. However, as noted above, comparisons of the EVA methods cannot be done by the mean of x_m. This issue does not concern the median. Therefore, the median of x is used as the measure of goodness in the following.

Monte-Carlo simulations were made to see how well the three EVA methods estimate the median of the variable value $x_{30/31} = F^{-1}(1 - 1/31)$ from fits to samples of size $N = 15$. In other words, we extrapolated from each individual fit to the 15 observations, obtained in Section 3.1, and the median of $x_{30/31}$ was calculated. These median values are shown in Table 4 for the five EVA methods. Next, 100,000 sets of 30 samples were generated by a Monte-Carlo simulation from the same selected distributions. For the largest values, at N = 30, the median was also determined, and is marked as 'From the distribution' in Table 4.

Table 4 indicates that the VWLS method proposed here is quite accurate in estimating the return value of an extrapolated extreme in terms of its median. In that regard too, the VWLS method clearly outperforms the Maximum Likelihood method for all three types of extreme value distributions. Table 4 also shows that PWM and L-moments perform similarly, and LH2-moments slightly worse than them. VWLS outperforms PWM, L and LH2 methods as well. This improved accuracy is very important in practical risk analysis.

3.4. Application example

The results of an application of the new method to an extensive analysis of real environmental flood-related data are presented in

Table 4
Median of the variable value $x_{30/31}$ determined from three extreme value distributions ($\alpha = 10$; $\mu = 70$) and by extrapolating from the fits obtained by the five EVA methods for N = 15.

Distribution type	From the distribution	VWLS	ML	PWM	L	LH2
Fréchet, $\gamma = 0.3$	**139.9**	139.4	125.7	120.7	120.8	118.9
Weibull, $\gamma = -0.3$	**92.6**	92.8	89.5	91.5	91.5	91.3
Gumbel	**107.7**	108.1	101.1	102.6	102.7	101.8

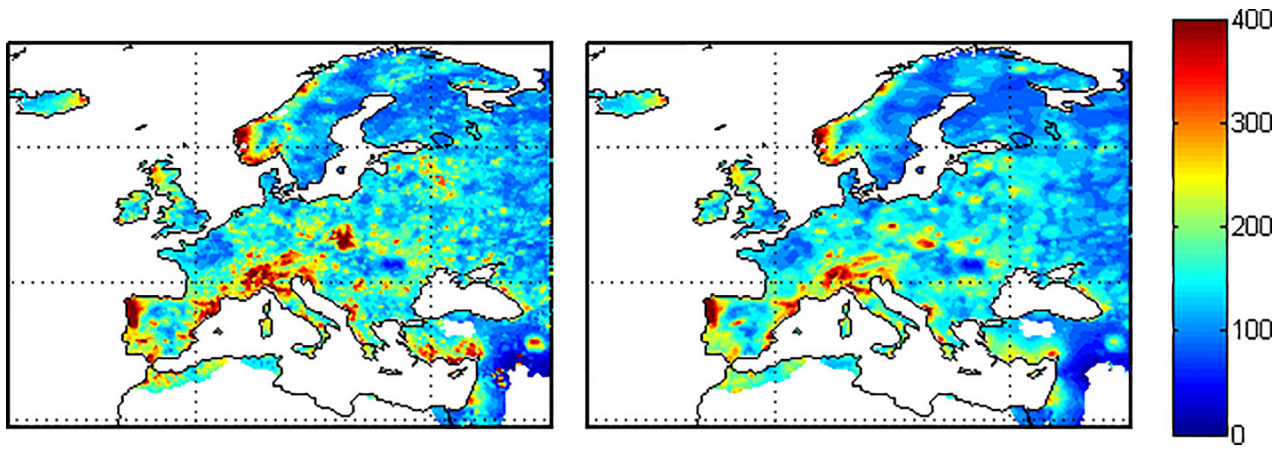

Fig. 4. Fifty-year return value of the annual maximum five-day precipitation total (in mm) based on the E-OBS data (Haylock et al. 2008) for the years 1971–2000. On the left, the data are analysed with the present VWLS method, and on the right, with the maximum likelihood method.

Fig. 4. Here, the fifty-year return value (the annual extreme that is exceeded once in 50 years in the mean) of the annual maximum five-day precipitation is calculated with both the proposed method (on the left) and the maximum likelihood method (on the right) based on 30 years of data of annual extremes. Such persistent heavy precipitation is a typical situation that precedes flooding events.

The underlying data are from the E-OBS European high-resolution land-only gridded dataset (Haylock et al., 2008) and includes 44,800 samples of 30 annual precipitation values in 1971–2000. In spite of this huge set of data, we found no numerical problems when applying the new method. The E-OBS dataset used here is produced through a spatial kriging interpolation of daily station data and computing gridded data at a 0.5° regular latitude-longitude grid. Details of the interpolation methods, their implementation and the calculation of uncertainties are given in Haylock et al. (2008).

Fig. 4 illustrates that our new VWLS method provides quite similar results compared to those obtained by the maximum likelihood method. Fig. 4 also indicates that VWLS is more sensitive to variations in the data, as it appears to provide more small-scale details that follow the local topography.

3.5. Conclusions

The VWLS method presented here provides estimates for the extremes that are considerably better than obtained by presently available EVA methods, particularly for small data sets. The main advantages of the new method are better prediction of the highest and smallest value in the data set, and of extreme events extrapolated to smaller probabilities than covered by the data. An additional benefit is that using this method requires no subjective methodological decisions by the user.

The weaknesses of VWLS method include the i.i.d. and the stationarity assumptions. These limitations may be removed upon further development and testing in a wider range of distribution parameters. The method is best suited for data originating from a single distribution. When this is not the case, this method can be expanded by a two-component approach (Rulfova et al., 2016). The method may be further improved by e.g. including in the weighing procedure the fact that order statistics are correlated.

Declaration of interests

None.

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References

Beirlant, J., Goegebeur, Y., Segers, J., Teugels, J., 2004. Statistics of Extremes. Wiley, Chichester.
Castillo, E., 1988. Extreme Value Theory in Engineering. Academic Press, New York.
Castillo, E., Hadi, A.S., Balakrishnan, N., Sarabia, J.M., 2005. Extreme Value and Related Models with Applications in Engineering and Science. Wiley, New Jersey.
Coles, S.G., 2001. An Introduction to Statistical Modeling of Extreme Values. Springer-Verlag, London.
Dixon, W.J., 1950. Analysis of extreme values. Ann. Math. Stat. 21, 488–506.
Fisher, R.A., 1922. On the mathematical foundations of theoretical statistics. Phil. Trans. Roy. Soc., Ser. A 222, 309–368.
Fisher, R.A., Tippett, L.H.C., 1928. Limiting forms of the frequency distributions of the largest or smallest members of a sample. Proc. Cambridge Phil. Soc. 24, 180–190.
Furrer, E.M., Katz, R.W., 2008. Improving the simulation of extreme precipitation events by stochastic weather generators. Water Resour. Res. 44, W12439.
Gilleland, E., Ribatet, M., Stephenson, A.G., 2013. A software review for extreme value analysis. Extremes 16, 103–119.
Greenwood, J.A., Landwehr, J., Matalas, N.C., Wallis, J.R., 1979. Probability weighted moments: Definition and relation to parameters of several distribution expressible in inverse form. Water Resour. Res. 15, 1049–1054.
Gumbel, E.J., 1958. Statistics of Extremes. Columbia University Press, New York.
Harris, R.I., 1996. Gumbel re-visited – a new look at extreme value statistics applied to wind speeds. J. Wind Eng. Ind. Aerodyn. 59, 1–22.

Haylock, M.R., Hofstra, N., Klein Tank, A.M.G., Klok, E.J., Jones, P.D., New, M., 2008. A European daily high-resolution gridded dataset of surface temperature and precipitation. J. Geophys. Res. (Atmospheres) 113, D201192008.
Hosking, J.R., 1985. Maximum-likelihood estimation of the parameters of the generalized extreme-value distribution. Appl. Stat. 34, 301–310.
Hosking, J.R.M., 1990. L-moments: analysis and estimation of distributions using linear combinations of order statistics. Roy. Stat. Soc. London 52, 105–124.
Hosking, J.R.M., Wallis, J.R., 1997. An Approach Based on L-Moments. Regional Frequency Analysis. Cambridge Univ. Press.
Hosking, J.R., Wallis, J.R., Wood, E.F., 1985. Estimation of the generalized extreme value distribution by a method of probability weighted moments. Technometrics 27, 251–261.
Jordaan, I., 2005. Decisions under Uncertainty. Cambridge Univ. Press.
Katz, R.W., Parlange, M.B., Naveau, P., 2002. Statistics of extremes in hydrology. Adv. Water Resour. 25, 1287–1304.
Landwehr, J.M., Matalas, N.C., Wallis, J.R., 1979. Probability weighted moments compared with some traditional techniques in estimating Gumbel parameters and quantiles. Water Resour. Res. 15, 1055–1064.
Lieblein, J., 1974. Efficient methods of extreme value methodology. Natl. Bur. Stand. (US) Rep. NBSIR, 75–602.
Makkonen, L., 2008a. Problems in the extreme value analysis. Struct. Safety 30, 405–419.
Makkonen, L., 2008b. Bringing closure to the plotting position controversy. Commun. Stat.–Theory Methods 37, 460–467.
Makkonen, L., 2006. Plotting positions in extreme value analysis. J. Appl. Meteorol. Climatol. 45, 334–340.
Makkonen, L., Pajari, M., 2014. Defining sample quantiles by the true rank probability. J. Prob. Stat. 326579. https://doi.org/10.1155/2014/326579.
Makkonen, L., Pajari, M., Tikanmäki, M., 2013. Discussion on Plotting positions for fitting distributions and extreme value analysis. Can. J. Civ. Eng. 40, 927–929.
Martins, E.S., Stedinger, J.R., 2000. Generalized maximum likelihood generalized extreme-value quantile estimators for hydrological data. Water Resour. Res. 36, 737–744.
McNeil, A.J., Frey, R., Embrechts, P., 2005. Quantitative Risk Management: Concepts, Techniques, and Tools. Princeton University Press, Princeton.
Pajari, M., Tikanmäki, M., Makkonen, L., 2019. Probabilistic comparison of quantile estimators for continuous random variables. Submitted to Comm. Stat. - Theory, Methods.
Pickands, J., 1975. Statistical inference using extreme order statistics. Ann. Stat. 3, 119–131.
Rasmussen, P.F., Gautam, N., 2003. Alternative PMW-estimators of the Gumbel distribution. J. Hydrol. 280, 265–271.
Reiss, R.-D., Thomas, M., 2007. Statistical Analysis of Extreme Values with Applications to Insurance, Finance. Hydrology and Other Fields, Birkhäuser, Boston.
Rulfova, Z., Buishand, A., Roth, M., Kysely, J., 2016. A two-component generalized extreme value distribution for precipitation frequency analysis. J. Hydrol. 534, 659–668.
Smith, R.L., 1989. Extreme value analysis of environmental time series: an application in trend detection of ground-level ozone. Statist. Sci. 4, 367–393.
Stephenson, A., Gilleland, E., 2005. Software for the analysis of extreme events: the current state and future directions. Extremes 8, 87–109.
Wang, Q.J., 1991. The POT model described by the generalized Pareto distribution with Poisson arrival rate. J. Hydrol. 129, 263–280.
Wang, Q.J., 1997. LH-moments for statistical analysis of extreme events. Water Resour. Res. 33, 2841–2848.